

Sage Research Methods Cases Part 2

A Practical Guide to Biological Primary Data Collection in an Impoverished Urban Setting: Illuminating Structural and Social Influences on Population Health Inequity

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Author: Jay A. Pearson, Arline T. Geronimus

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Abstract

Integrating knowledge from the molecular and social sciences has become an important avenue toward understanding how social experience becomes biologically embedded to affect health adversely over the life course in marginalized populations. Conducting rigorous and deeply integrated biopsychosocial research is complicated by practical and ethical challenges. In particular, identifying and successfully recruiting study participants from marginalized communities for sociological research poses unique challenges, and these challenges are exacerbated when the research efforts include collection of biological samples. Our dynamic and contingent theoretical understanding of how socially structured experiences become biologically embedded to activate population health inequities—the weathering hypothesis—led us to focus this study on a single economically depressed city, Detroit, Michigan, rather than a national sample and to apply community-based public health principles to collect biological data from residents. We found linking to an established community-based partnership between our university and Detroit community partners was the first, but far from the last step to successfully collecting biological specimens from a diverse, low-income, and historically exploited set of participants. The purpose of this case study is to share the lessons learned from our research experience and provide a set of practical guidelines for collecting biological data from an ethnically and racially diverse economically depressed urban community.



Learning Outcomes

By the end of this case, students should be able to

- Learn methods for constructing functional community-based primary data collection projects
- Learn how to anticipate, recognize, articulate, and address unique challenges associated with primary data collection from hard to reach populations
- Learn the value of utilizing constructs from community-based participatory research and cultural humility to develop trust and gain entry while actively engaging a community of interest
- Learn how to develop respectful community engagement strategies appropriate for ethnically diverse populations

Project Overview and Context

Individual behavioral and socioeconomic explanations account for only a portion of the systematic social disadvantages and health inequities characterizing the lives of impoverished urban residents of color ([Pearson, 2008](#)). Holding greater promise toward elucidating the underlying causes of persistent health disadvantage are investigations which explore the physical embodiment and biological embedding of lived social experience by identity category. One line of inquiry to address these mechanisms proposes to explore how being racialized affects population health inequity.

Integrating knowledge from the molecular and social sciences has become an important avenue toward understanding how social experience becomes biologically embedded to affect health adversely over the life course in marginalized populations. Conducting rigorous research that can be interpreted to good scientific and policy purpose is complicated by practical and ethical challenges as well as by limitations in expertise faced by biological and social scientists who hope to collaborate together or at least to integrate data from fields in which they are not trained into their own data sets. A common approach has been to augment existing national health or social surveys with biological data. Although understandable and expedient, this approach is ultimately incomplete. Often in big national surveys biological data are collected without regard to theory. Moreover, given the nature of national survey sampling, such surveys are underpowered for the study of the most marginalized, those at the intersections of long-term poverty, oppressed race/ethnicity, or residence in stigmatized, high-poverty places, whether in Appalachia or impoverished urban areas. In national samples, the poorest Whites are disproportionately rural, making estimates of the size of racial or ethnic differences in health subject to serious residual confounding. Neighborhood analyses of national data sets often rely on census administrative units or test for independent effects using a small number of setting characteristics rather than viewing neighborhoods holistically as places where meaning to residents may be structured by historical factors, ideologically informed expectations, and geographically rooted social ties. When taking this common risk factor approach to the social determinants of health, statistical correlations to a measured aspect of a setting or a racial-ethnic/socioeconomic group may obscure the more complex underlying processes that shape the observed patterning of population health ([Geronimus, 2001](#); [Pearson, 2008](#); Reed & Chowkanyen, 2011). In particular, the impacts of uncommon intersectional identities—for example, those of high-poverty urban Whites—are easily lost. This loss silences their voices and obscures their needs. It also hampers any scientific ability for accurate and nuanced understanding of how race, income, and place interact to produce health disadvantages for populations of color. In addition, if population health over the life course is sensitive

to life experience and environmental context, the assumption of national sampling—that a randomly selected low-income White, Black, or Latino is representative of all comparably low-income Whites, Blacks, or Latinos—cannot be taken as axiomatic across time and place.

To begin to address the reviewed shortcomings of health research seeking to integrate biological and sociological data, we chose to focus on a single economically depressed city, Detroit, Michigan. Detroit is characterized by a high degree of race-based residential segregation and associated economic disinvestment. Its budget problems became so severe that, in 2013, the city of Detroit filed for Chapter 9 bankruptcy—the largest municipal bankruptcy filing in U.S. history by debt—estimated at US\$18 to 20 billion—or by population size. It has suffered deeply from “austerity urbanism” (Peck). Population outmigration and economic divestment over the past five decades have reduced Detroit’s population from 1.8 million in 1950 to just 706,585 in 2011, only 10% of whom are White. Outmigration of Whites has been especially severe, suggesting that the Detroit White population is a distinctively disadvantaged White population. Studies investigating the health of impoverished urban populations have most frequently been restricted to Black Americans and select Latino Sub-Populations of Puerto Ricans and Mexican origin ([Henry-Sanchez & Geronimus, 2013](#)). The current investigation offered the unique opportunity to examine the immediate social environment and health of three distinct racial and ethnic populations, Black Americans, White Americans, and Latinos, similarly situated in an economically depressed mid-western city. To optimize the chance that sociological theory structured our research design, data analysis, and interpretation of findings—rather than risk turning to biological folk notions of the genomic era or understanding of urban contexts propagated in the power literature—we developed our hypotheses drawing on The “Weathering Hypothesis.” Weathering is a comprehensive conceptual framework for theorizing how social processes of racialization are maintained by the dominant culture and expressed in the unequal distribution of power, material resources, identity affirmation, and requirements for persistent high effort practical and psychosocial coping with the environmental and policy consequences of racialization, and in turn, how these together affect population health differentially across racialized populations and cumulatively across the life course to accelerate the development and progression of chronic disease, increase health vulnerability, and ultimately accelerate aging in marginalized groups ([Geronimus, 1992](#); [Geronimus, Hicken, Keene, & Bound, 2006](#)). At the time of its conception, the absence of theoretically informed or dynamic molecular measures limited rigorous investigation to testing hypothesized relationships implied by the Weathering Hypothesis between social factors and specific disease outcomes such as hypertension and poor birth outcomes. Since then two promising bio-measures, allostatic load (AL), a measure of stress-mediated wear and tear on important biological systems—including the cardiovascular, the metabolic, and the immune

systems—and telomere length (TL), a molecular measure of biological aging, have emerged. We sought to test hypotheses following from the weathering hypothesis that variations in exposure to environmental and psychosocial stressors among Detroit Black, Latino, and White residents would influence population levels of AL and affect population TL. To measure TL requires venous blood collection and DNA extraction. To measure AL requires blood and saliva collection. The remainder of this case study documents our efforts to collect primary biological and sociological data from a racially and ethnically diverse urban setting.

We applied community-based public health principles and worked with an already existing community based effort the Healthy Environments Partnership (HEP) to collect biological data from residents identified by their ongoing community-based research. Working with HEP offered us entry into the community and contact with potential participants whose biological data could be linked to detailed personal, community, social, and environmental data at a more fine-grained level than national surveys often provide. HEP had launched two waves of a community survey designed to examine and address aspects of Detroit's social and physical environment and their effects on residents of three neighborhoods ([Schulz et al., 2005](#))

Research Practicalities

The Detroit civil rights protests of the 1960s sparked a marked increase in the number of external researchers and projects exploring urban sociological phenomena within the city ([Warren, 1973](#)). This focus on Detroit has continued until the present. Consequently, and not surprisingly, many Detroit residents suffer from research burnout and have developed a justified mistrust of university scientists ([True et al., 2017](#)).

This reality of contentious university community relations combined with outmigration of relatively well-to-do residents, public economic divestment, job loss, urban blight, and austerity policies has resulted in profound and persistent social distress running parallel to a justified mistrust of researchers purporting commitment to improve conditions for Detroit residents. There exists no single social scientific model which adequately addresses these challenges while simultaneously providing guidance toward the successful identification and recruitment of study participants in a way that is respectful, unobtrusive, and non-coercive. At the crux of this challenge is the very real specter of Detroit community members viewing researchers with skepticism.

Our task, soliciting primary and biological data from a hard to reach disproportionately transient diverse urban population, was further exacerbated by an established and relatively well-known history of researchers taking

up temporary residence in Detroit for time-limited projects extracting data from communities while offering little in return. Widespread and legitimate concerns about volunteering biological specimens to research staff at an elite public university were fueled both by the unethical treatment of Black Americans in widely publicized and relatively recent public health research—most notoriously, through the Tuskegee syphilis study—and a continuing history of social scientists construing the health problems of Blacks as essential either to their DNA or to their culture and behavior and, in these molecularized times, to the effects on their DNA of their culture and behavior.

The community-based participatory research (CBPR) framework places emphasis on long standing, well-established relationships which prioritize the perspectives of community residents and served as an excellent mechanism to engage our communities of interest. For CBPR to be successful, community members and organizations share in developing and conducting all aspects of the research endeavor as coequal partners with university researchers ([Belone et al., 2016](#); [Israel et al., 2010](#); [Minkler & Wallerstein, 2008](#); [Schulz et al., 2005](#)). We found that taking a CBPR approach facilitated entry into the community; however, we found ourselves facing a range of other structurally rooted impediments to data collection that we had to address for biological specimen collection to be successful. These ranged from practical issues related to deep poverty that fueled logistical problems, to discomfort among some members of our project staff in engaging Detroit residents, as well as mistrust of us on the part of Detroit residents, all arguably borne of a history of structural racism.

Research Design

In 2002-2003 and again in 2008-2009, the Healthy Environment Partnership (HEP), a CBPR effort designed to examine and address aspects of the social and physical environment in three Detroit neighborhoods, fielded two waves of a community survey in Detroit, consisting of a stratified, multi-stage probability sample of residents aged 25 and older living in three geographic communities within the city of Detroit (Northwest, Southwest, and Eastside). Through collaboration with HEP, our project proposed blood and saliva collection to participants in the second wave of the HEP survey and subsequent analysis for TL and AL and linkage to their community survey responses.

In partnership with HEP staff, we developed a set of standard procedures to follow up with respondents from Wave 2 of the HEP survey. At the conclusion of Wave 2 interviews, each respondent was told of our weath-

ering study and asked whether they would be interested in participating. This follow-up to the Wave 2 survey involved a venous blood draw and collecting 3 days of saliva samples. Residents agreeing to participate in this component of the study completed a separate written informed consent form and were added to a list of participants to be contacted and scheduled for a blood draw to be conducted at a local community center, mobile clinic or in the residents' home.

Willing participants were called by the project director who used a HEP and UM Institutional Review Board (IRB)-approved protocol script to schedule a clinic appointment or a home visit with a phlebotomist to draw venous blood according to standard procedures of blood collection. During this phone call, the participant was read a copy of the consent form and asked to confirm their home address so that they could be mailed the saliva sample kit and instructions. Upon receipt of the saliva sample collection kit, the participant was responsible for collecting three saliva samples per day for 3 days scheduled to end the night before blood collection. Blood was drawn on the morning subsequent to a 12-hr fast and abstention from smoking and a 24-hr abstention from alcohol. All blood collection tubes and accompanying paperwork were labeled with assigned identification numbers only (no personal identifiers).

Of the 460 Wave 2 survey participants, 260 (56.5%) expressed interest in the weathering study. All 260 individuals were later contacted by the study coordinator and given more information regarding the purpose of the study, participation requirements, and honorarium for participating. After completing an informed consent process, 239 (92% of those asked) enrolled in the weathering study, completed a brief in-person interview, and provided saliva samples (5%), blood samples (1%), or both saliva and blood samples (94%) for analysis.

The following description outlines our *planned* data collection strategy developed in collaboration with and approved by the CBPR staff. Clinic appointments were scheduled for Saturday mornings at a local community center located in one of the three Detroit neighborhoods where the project was being conducted or at the University of Michigan research center in downtown Detroit. Study participants were assigned appointment times at approximately 10-min intervals during a 3-hr clinic session. A maximum of 18 participants could be scheduled for a single 3-hr clinic session. Clinic sessions started promptly at 8:00 a.m. and ended at 11:00 a.m. During the clinic sessions, two phlebotomists collected blood samples and two research assistants along with the project coordinator conducted participant interviews, reviewed the informed consent procedure, and collected saliva samples. A research assistant transported all collected blood and saliva samples, typically six sets, to the University lab at 9:00 a.m. and 10:00 a.m. The project coordinator transported the final set of samples to the lab at 11:00 a.m. At the community centers, we never had more than six participants scheduled for

a Saturday clinic, whereas at the university research center, we maximized our capacity of 18 participants on multiple occasions.

Residents who agreed to participate in the Weathering study but were unable to attend scheduled weekend clinic sessions were scheduled for a sample collection home visit. The project director and a single phlebotomist visited the homes of qualified residents between 8:00 a.m. and 9:30 am on weekday mornings. Due to time constraints associated with transporting blood samples from Detroit to the university processing lab, we anticipated a maximum of three home visits could be scheduled in a single day. The project director administered the informed consent, conducted the survey instrument, paid the honorarium, and delivered all samples to the processing lab.

Processed samples shipped to labs were sent using FedEx Overnight services. These shipments adhered to the requirements set forth by FedEx, the receiving labs, and the U.S. Government for the shipment of blood products.

Finally, all participants were mailed a health assessment feedback letter. Those who had biological assay values which warranted medical attention or were amenable to behavior change were notified in the feedback document.

Community Engagement

The success of our community engagement efforts was contingent upon developing a working partnership with staff from the HEP community-based partnership and benefited from professional relationships that existed prior to our project's formal launch. The HEP partnership was an extension of the Detroit Community Academic Urban Research Center which was established in 1995 with the express purpose of promoting collaborative CBPR that improves the health of Detroit residents. HEP involves collaboration among UM SPH, the Detroit Department of Health and Wellness Promotion, Brightmoor Community Center, Detroit Hispanic Development Corporation, Friends of Parkside, Henry Ford Health System, Rebuilding Communities Inc., and Southwest Solutions. HEP adopted a set of CBPR principles that include an emphasis on conducting research (basic or applied) that is beneficial to the community involved; including community, practitioner, and academic partners in all major phases of the research process; strengthening collaboration among all partners involved; enhancing the capacity of all partners; and disseminating research findings to community

members in ways that are understandable and useful.

The HEP team invited members of our project to community meetings, introduced us to key community leaders and to staff at the community centers where we conducted clinics. This facilitated entry and lent credibility to our efforts by validating our project as a continuation of well-established, existing programs familiar to many residents.

By collaborating with HEP and augmenting the biomeasure data in the central HEP database, we were in a unique position to estimate racial/ethnic and socioeconomic differences in weathering or biological aging among urban residents and also to gain purchase on the extent to which personal stressors, features of the social or physical environment, health behaviors, and multilevel socioeconomic indicators might mediate or moderate the relationship between race/ethnicity and cellular aging or AL. Finally, we were also able to provide new evidence on the biological mechanistic pathways through which cellular aging may occur and explore its relationship to the early onset of chronic disease.

Method in Action

Our project protocols endeavored to anticipate and account for dynamic contingencies associated with community-based primary data collection. Despite our best efforts, a number of unforeseen challenges quickly and regularly emerged. These challenges were of a dual nature.

The first set of challenges involved developing the most effective strategies to reach the population of interest, communicate the core mission and importance of the project and its value to the community. The second set of challenges involved negotiating between the needs and wants of a large research institution and respectful engagement of a community with a long history of research from the institution. The list below documents each individual major challenge:

1. The nature of an impoverished highly transient urban population made phone calls an ineffective communication method for HEP participants who agreed to participate in our study. Approximately 25% of interested participants had non-functional phone numbers. This was due to a combination of temporary residence and the use of “burners,” inexpensive short-term disposable phones.
2. Despite phone call reminders and mailed sample collection reminders, many residents either did not understand or fully appreciate the importance of maintaining fidelity to biological sample data collec-

tion protocols. This resulted in approximately 15% of early saliva collection samples having to be re-collected.

3. During home visits, the time required to administer the survey instrument and the written informed consent limited sample collections to 1 per day.
4. Cultural expectations associated with communication style and social engagement varied by identity category of the participant. Perceptions of what constituted a respectful interaction in the community of interest were contingent upon race, ethnicity, national origin, gender, socioeconomic status (SES), and neighborhood.
5. Requests from and expectations to change protocols or practices on the part of community residents and project staff were not always easily accommodated by the university or CBPR structures.
6. Staff turnover at the university processing lab was more frequent than anticipated. We recruited, hired, and trained three processing lab technicians over the course of our project. Each of these new hires required approximately 4 weeks of training and one of them left immediately after receiving the training.
7. Our project staff also suffered sudden turnover at the critical juncture of beginning actual data collection. Such departures surely had multiple causes. However, the precise timing of the departures, just as data collection was to become the central activity of the project among staff who had proven their dedication to the principles of the study in theory, suggested trepidation at the thought of actually interacting with Detroit community residents and the difficult logistical and possibly interpersonal requirements involved.

Responses to these challenges were varied and innovative. The project director realized early in the data collection process that he would have to spend considerably more time in Detroit than was initially anticipated. This time investment served several key purposes. First, residents with non-functioning phones were best located and recruited by home visits or community outreach. Second, many residents were more likely to successfully follow data collection protocols, particularly saliva sample collection, if they were visited by the project director who demonstrated for them exactly what was required and was available to answer extemporaneous questions. Third, a face to face reminder of upcoming clinic appointments or home visits for sample collection was significantly more effective than attempted phone calls or mailed letters. Fourth, early in the project, several participants who were scheduled for home visits expressed frustration with the time required to read and sign the informed consent, complete the survey instrument, and have blood samples collected during the visit. Participants suggested a preference for completing documents during the reminder visit. Amendments for changes to honor each of these suggestions had to be submitted to and approved by

the CBPR advisory board or the university IRB, as appropriate, before they could be implemented.

Understanding of and appreciation for the cultural context of community residents was essential for the successful and respectful collection of data. The project necessitated staff who were conversant in the cultural milieu of urban Black American, foreign, and U.S.-born Latino, and impoverished White American communities. Key to the success of the data collection effort was the fact that the project director occupies the intersectional social identity of a Spanish-speaking Black American male who has spent more than 25 years working in multi-cultural settings, including the rural south, Latin America, and northern urban communities. He possessed a unique combination of social and cultural skills that allowed him to respectfully engage members of the three communities. These qualities helped maintain trust of the project on the part of the participants and also gave the responsibility of interacting with them to a person who was not crippled by stress-inducing racialized phantasms. Each of the project field staff, two phlebotomists, and two research assistants were Black American women all of whom lived in one of the three Detroit communities.

Socio-cultural expectations and interactions for respectful individual and community engagement varied systematically by racial, ethnic, and national origin identity. The concept of cultural humility was incorporated into community engagement and guided the director's regular interactions with study participants. Cultural humility incorporates a personal commitment to self-evaluation and self-critique, to redressing the power asymmetries characterizing the researcher-participant dynamic, and to developing mutually beneficial and non-paternalistic partnerships with communities on behalf of individuals and defined populations (Arhienbuwa et al., 2000; [Tervalon & Murray-Garcia, 1998](#)). The following assessments provide a set of general though not universal or deterministic observations of these phenomena.

Black Americans had the most questions about a history of exploitive research practices in disadvantaged/marginalized communities. These included but were not limited to the infamous Tuskegee Syphilis study. Black Americans also expressed the greatest interest in the personal family history of the project coordinator. Project staff needed to be prepared for and willing to put in the time to address these questions.

Latinos, U.S., and foreign-born English and Spanish speaking were particularly interested in the ethnic and national origins of the project director. In addition, recruitment and reminder visits to Latino homes were most respectfully and effectively viewed as social engagements which typically required a more flexible time and personal investment than those of the White or Black American participants.

White Americans had the most questions about the professional qualifications of the staff and their capacity

to complete tasks associated with the sample collection procedures and the broader project. The sample collection dynamic was typically formal, direct, and occasionally terse.

The project director quickly realized that visiting residents for recruitment and reminders of clinic or home data collection appointments was an important tool in establishing credibility, fostering trust, encouraging accountability, and soliciting important information. Once the project director spent more time in the community and implemented this more culturally flexible approach, the likelihood that participants correctly completed saliva sample collection, adhered to other clinic prep protocols, and arrived on time for scheduled sample collection all saw improvements.

Ultimately, community outreach proved to be a more valuable time investment for success of the project than on campus office tasks. This posed a challenge given the limited available staff and suggests the critical importance of budgeting for a larger staff than we had anticipated needing based on our experience with previous projects collecting data from suburban or student populations.

Practical Lessons Learned

Effectively communicating the value of following sample collection protocols was the single greatest challenge and necessitated changes in both the initial direction and the follow-up. This required regular community visits to establish rapport, answer questions, demonstrate and explain the importance of adhering to protocols. Related to this was the realization that hand delivering saliva sample collection kits increased the likelihood that the samples would be collected correctly. The labor and time-intensive nature of participant location, recruitment, sample collection kit delivery, appointment reminders, and follow-up competed with the fulfillment of daily office tasks, data entry, and other campus responsibilities. The number of staff at the beginning of project was inadequate to accommodate the flexible and in-person approach that we found was needed to collect biological specimens in the face of legitimate and structured distrust.

We found linking to an established community-based partnership between our university and Detroit community partners was the first, but far from the last step to successfully collecting biological specimens from a diverse, low-income, and historically exploited set of participants. Cultural humility, social engagement, and dynamic innovation were essential to the success of our community-based primary data collection effort. We also learned that assumptions about concepts of community should be avoided. Discovering over time how

community members view and define concepts of their own identity is the preferred method. For example, in concert with our own presumptions and the views of community partners, we had initially planned to collect data at local community centers. We erroneously thought they would be the ideal location because they were in close proximity to the participants' homes. However, contrary to our expectations, many respondents regardless of community of residence preferred the centrally located university research center. This prompted the project director to set aside assumptions about the value of geographically defined community boundaries and preferences associated with them, deferring instead to those dimensions of identity most salient and consequently appropriate for study participants.

Conclusion

In an increasingly diverse nation characterized by great and increasing structural, social, and economic inequality and entrenched socioeconomic and racial/ethnic residential segregation, the value of frameworks and modalities which effectively guide solicitation of data and information from hard to reach communities and populations is imperative.

Although CBPR was excellent for facilitating entry into the community of interest and lending credibility to our efforts, it did not lend itself readily to the daily demands of dynamic and rapidly evolving contingencies associated with primary biological data collection from socially disadvantaged populations. A component of the CBPR approach, cultural humility, and deference to the expertise of community members was, however, essential to our success. In addition, insights provided by project staff who lived in the community of, but were not themselves a part of CBPR proved to be an invaluable resource. In summary, the greatest challenge was devising a strategy that allowed the project team to simultaneously incorporate suggestions from knowledgeable staff while maintaining fidelity to the project plan and respect for community members volunteering as participants for our study.

Exercises and Discussion Questions

1. What role might identity of the researcher play in primary data collection with hard to reach populations?
2. How do researchers account for the influence of historical events in supporting or compromising pri-

mary data collection efforts?

3. What major challenges might community university partnership efforts expect to encounter?
4. Offer suggestions for how researchers not native to communities of interest might come to understand concepts of “community” shared by residents.

Further Reading

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Web Resources

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